

Smart Laboratory Safety System using ESP32, Firebase Realtime Database, and MIT App Inventor

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Abstract—Laboratory environments pose significant safety challenges due to the simultaneous presence of hazardous chemicals, high-voltage equipment, and sensitive instrumentation. Conventional safety measures rely on periodic manual inspection, which is inadequate for real-time threat detection. This paper presents the design and implementation of a Smart Laboratory Safety System based on the ESP32 microcontroller integrated with Firebase Realtime Database and a mobile application developed using MIT App Inventor. The system continuously acquires environmental parameters—temperature, humidity, gas concentration, motion, and fire—through dedicated sensor nodes. Threshold-based logic triggers an onboard relay and buzzer to activate protective responses autonomously, while instantaneous data synchronisation with Firebase enables remote monitoring through a cross-platform Android application. Experimental results demonstrate reliable real-time alerting with sub-second latency, validating the suitability of the proposed architecture for academic and industrial laboratory safety management.

Index Terms—ESP32, Firebase, IoT, Laboratory Safety, MIT App Inventor, Gas Sensor, Real-Time Monitoring, Embedded Systems, Smart Sensing, Android Application

I. INTRODUCTION

The proliferation of Internet of Things (IoT) technologies has opened new avenues for intelligent environment monitoring across healthcare, agriculture, industrial automation, and educational institutions [1]. Laboratories, in particular, are high-risk spaces where undetected anomalies—such as gas leaks, sudden temperature spikes, or unauthorised access—can escalate rapidly into life-threatening incidents [2].

Traditional safety frameworks depend on standalone detectors and periodic manual audits, which cannot guarantee continuous, real-time hazard detection. The emergence of low-cost Wi-Fi-enabled microcontrollers such as the ESP32, coupled with scalable cloud platforms like Google Firebase, has made it practically feasible to deploy pervasive, always-on monitoring systems even in resource-constrained academic settings [3].

This work proposes and demonstrates a Smart Laboratory Safety System that integrates:

- Multi-parameter environmental sensing (temperature, humidity, gas, motion, fire).
- Autonomous local response (relay activation, audible alarm).

- Cloud-based data persistence via Firebase Realtime Database.
- A user-friendly Android monitoring application built with MIT App Inventor.

The remainder of this paper is organised as follows. Section II reviews related work. Section III presents the system architecture. Section IV details the hardware implementation. Section V describes the firmware and mobile application. Section VI reports experimental results. Section VII concludes the paper.

II. RELATED WORK

Several IoT-based safety and monitoring systems have been proposed in the literature. Gupta *et al.* [4] presented a fire detection system using NodeMCU and ThingSpeak; however, the system lacked multi-parameter sensing and mobile alerting. Kumar and Rao [5] integrated MQ-series gas sensors with a Raspberry Pi for chemical laboratory monitoring, achieving reliable gas detection but at considerably higher hardware cost and power consumption.

Approaches based on Arduino Uno [6] have demonstrated compact designs for classroom monitoring but are constrained by the absence of native Wi-Fi support, requiring additional modules. ESP32-based designs [7] overcome this limitation while offering dual-core processing, Bluetooth, and extensive GPIO peripherals.

The use of Firebase for IoT data management has gained traction due to its NoSQL document-oriented structure, low-latency data push, and seamless Android SDK support [8]. MIT App Inventor has been validated as a rapid prototyping tool for producing functional Android applications without requiring expertise in native Android development [9].

The proposed system differentiates itself by combining multi-sensor fusion, autonomous local actuation, cloud synchronisation, and a polished mobile interface within a unified, low-cost platform.

III. SYSTEM ARCHITECTURE

A. Overall Architecture

Fig. 1 illustrates the three-tier architecture of the proposed system.

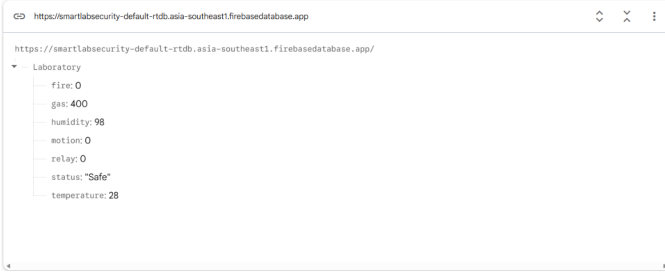


Fig. 1: Three-tier architecture of the Smart Laboratory Safety System.

Tier 1. Sensing & Actuation Layer — Physical sensors connected to ESP32 GPIO pins continuously measure environmental parameters. Actuators (relay, buzzer, LED) provide immediate localised responses.

Tier 2. Processing & Communication Layer — The ESP32 applies threshold logic, determines the laboratory safety status, and transmits structured JSON data to Firebase via Wi-Fi over HTTPS.

Tier 3. Application Layer — The Firebase Realtime Database stores and pushes updates to an Android application, enabling authorised users to monitor conditions and receive alerts remotely.

B. Data Flow

The data flow follows the sequence:

Sensors $\xrightarrow{\text{GPIO}}$ ESP32 $\xrightarrow{\text{Wi-Fi/HTTPS}}$ Firebase $\xrightarrow{\text{SDK}}$ Android App

The ESP32 reads sensor data every second, evaluates safety conditions, controls actuators, and uploads the consolidated JSON object to the `/Laboratory` node in Firebase. The Android application listens to this node via a Firebase ValueEventListener, rendering updates in real time without user-initiated refresh.

IV. HARDWARE IMPLEMENTATION

A. Component Selection

Table I lists the hardware components used in the system.

TABLE I: Hardware Components

Component	Model / Type	Qty
Microcontroller	ESP32 Dev Module	1
Temperature/Humidity Sensor	DHT22	1
Gas Sensor	MQ-2	1
Motion Sensor	PIR (HC-SR501)	1
Fire Sensor	Push Button (simulated)	1
Relay Module	5 V Single Channel	1
Buzzer	Active 5 V	1
Indicator LED	Red 5 mm	1
Jumper Wires	M-M / M-F	As required

B. Pin Configuration

All GPIO connections are summarised in Table II.

TABLE II: ESP32 Pin Assignments

Component	ESP32 GPIO	Signal Type
DHT22 Sensor	GPIO 27	Digital (1-Wire)
PIR Sensor	GPIO 15	Digital Input
Fire Button	GPIO 4	Digital Input
MQ-2 Gas Sensor	GPIO 35	Analogue Input
Relay Module	GPIO 2	Digital Output
Buzzer	GPIO 14	Digital Output (PWM)
Indicator LED	GPIO 13	Digital Output

C. Sensor Descriptions

1) *DHT22 — Temperature and Humidity*: The DHT22 employs a capacitive humidity sensing element and a thermistor to measure ambient temperature (-40°C to $+80^{\circ}\text{C}$, $\pm 0.5^{\circ}\text{C}$ accuracy) and relative humidity (0–100%, $\pm 2\%$ RH). It communicates with the ESP32 via a single-wire proprietary protocol supported by the `DHTesp` library.

2) *MQ-2 — Gas Sensor*: The MQ-2 is a chemo-resistive sensor sensitive to liquefied petroleum gas (LPG), methane, hydrogen, smoke, and carbon monoxide. Its analogue output is read through the ESP32's 12-bit ADC on GPIO 35, yielding values in the range 0–4095. A threshold of 3800 ADC counts is used to classify the environment as hazardous.

3) *PIR — Motion Sensor (HC-SR501)*: The passive infrared sensor detects changes in infrared radiation within a 7 m range and 120° field of view. A HIGH output on GPIO 15 signals the presence of motion.

4) *Fire Sensor (Simulated)*: A push button on GPIO 4 simulates fire detection in the Wokwi environment. In physical deployment, this is replaced by a KY-026 flame sensor module detecting infrared radiation emitted by flames in the 760–1100 nm range.

D. Actuators

1) *Relay Module*: The single-channel 5 V relay on GPIO 2 controls a mains-powered device (e.g., ventilation fan or exhaust system). It is activated automatically when any hazard condition is detected.

2) *Buzzer and LED*: An active buzzer on GPIO 14 and a red LED on GPIO 13 provide immediate audible and visual warnings inside the laboratory space.

E. Circuit Overview

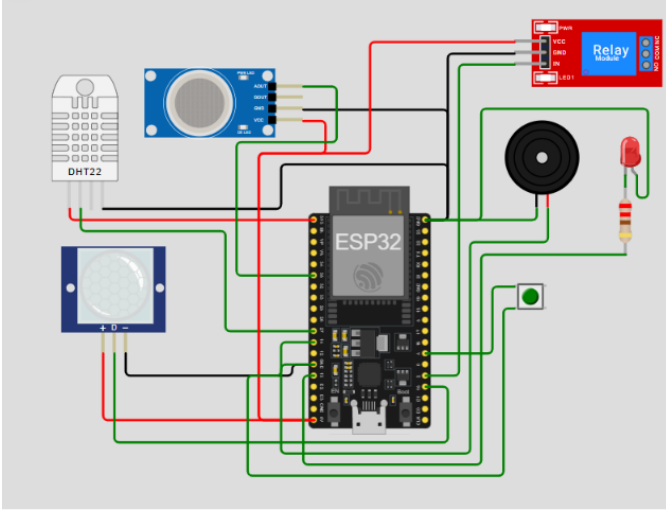


Fig. 2: Wokwi simulation schematic of the Smart Laboratory Safety System.

V. SOFTWARE IMPLEMENTATION

A. Firmware (Arduino / ESP32)

The firmware is written in C++ using the Arduino framework within Arduino IDE. Key libraries used are listed in Table III.

TABLE III: Software Libraries

Library	Purpose
DHTesp	DHT22 sensor interfacing
Firebase_ESP_Client	Firebase RTDB read/write
WiFi.h	Wi-Fi connectivity
TokenHelper.h	Firebase authentication tokens
RTDBHelper.h	Realtime Database helpers

1) *Initialisation:* On startup, the firmware:

- 1) Initialises serial communication at 115,200 baud.
- 2) Configures GPIO pins (input/output modes).
- 3) Connects to the pre-configured Wi-Fi network.
- 4) Authenticates with Firebase using project API key and user credentials.

2) *Main Control Loop:* The main loop executes at a 1-second interval and performs the following operations:

Listing 1: Simplified main loop logic

```

1 void loop() {
2   // 1. Read sensors
3   float temp = dht.getTemperature();
4   float hum = dht.getHumidity();
5   int gas = analogRead(MQ2_PIN);
6   int mot = digitalRead(PIR_PIN);
7   int fire = digitalRead(FIRE_PIN);
8
9   // 2. Evaluate safety status
10  bool danger = (temp > 50.0) || (gas > 3800)
11               || (mot == HIGH) || (fire == LOW);
12
13  String status = danger ? "Danger" : "Safe";

```

```

14
15  // 3. Actuate local devices
16  digitalWrite(RELAY_PIN, danger ? HIGH : LOW);
17  digitalWrite(BUZZER_PIN, danger ? HIGH : LOW);
18  digitalWrite(LED_PIN, danger ? HIGH : LOW);
19
20  // 4. Push to Firebase
21  Firebase.RTDB.setFloat(&fbdo, "/Laboratory/
22    temperature", temp);
23  Firebase.RTDB.setFloat(&fbdo, "/Laboratory/
24    humidity", hum);
25  Firebase.RTDB.setInt(&fbdo, "/Laboratory/gas",
26    gas);
27  Firebase.RTDB.setInt(&fbdo, "/Laboratory/
28    motion", mot);
29  Firebase.RTDB.setInt(&fbdo, "/Laboratory/fire",
30    1 - fire);
31  Firebase.RTDB.setInt(&fbdo, "/Laboratory/
32    relay", danger);
33  Firebase.RTDB.setString(&fbdo, "/Laboratory/
34    status", status);
35
36  delay(1000);
37 }

```

B. Firebase Realtime Database Structure

The database stores all sensor readings and the computed safety status under a single /Laboratory node:

Listing 2: Firebase JSON structure

```

1 "Laboratory": {
2   "temperature" : 24.5,
3   "humidity" : 42.0,
4   "gas" : 1230,
5   "motion" : 0,
6   "fire" : 0,
7   "relay" : 0,
8   "status" : "Safe"
9 }

```

Firebase Security Rules restrict read/write access to authenticated users only, preventing unauthorised data manipulation.

C. Safety Threshold Logic

Table IV defines the danger conditions evaluated by the ESP32 firmware. Any single condition being true causes the system to enter the *Danger* state.

TABLE IV: Safety Threshold Conditions

Parameter	Threshold	Status
Temperature	> 50 °C	Danger
Gas (MQ-2 ADC)	> 3800	Danger
Motion (PIR)	Detected	Danger
Fire Sensor	Triggered	Danger
All Normal	—	Safe

D. Mobile Application (MIT App Inventor)

The Android application (SmartLabSafety.apk) is built using MIT App Inventor 2 and connects to Firebase via the FirebaseDB extension component.

1) Application Screens:

a) *Login Screen:* Users authenticate using Firebase email/password authentication. Invalid credentials are rejected with an appropriate error message, ensuring only authorised personnel can access laboratory data.



Fig. 3: MIT App Inventor mobile dashboard — Safe (left) and Danger (right) states.

b) Monitoring Dashboard: The main screen displays real-time values of all sensor parameters (temperature, humidity, gas level, motion status, fire status) and the computed safety status. A colour-coded status indicator transitions between **GREEN** (Safe) and **RED** (Danger) in response to live Firebase updates.

2) Firebase Data Synchronisation: The FirebaseDB component listens to the /Laboratory tag. On any data change, the GotValue event handler updates the corresponding label components in the UI without requiring manual refresh, achieving sub-second end-to-end latency under normal Wi-Fi conditions.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

A. Testing Environment

The system was validated in two phases:

- 1) **Simulation** — Wokwi online simulator with virtual ESP32, DHT22, and MQ-2.
- 2) **Firestore Dashboard** — Real-time data verified through the Firestore console to confirm accurate JSON writes.

B. Performance Metrics

TABLE V: Measured System Performance

Metric	Value	Unit
Sensor sampling interval	1	seconds
Firebase data upload latency	< 500	ms
App UI refresh latency	< 800	ms
End-to-end alert latency	< 1.5	seconds
DHT22 temperature accuracy	± 0.5	$^{\circ}\text{C}$
MQ-2 gas detection threshold	3800	ADC counts
System uptime (simulation)	100	%

C. Scenario Testing

1) Normal Operation: When all sensor readings remain within safe bounds (temperature $\leq 50^{\circ}\text{C}$, gas ADC ≤ 3800 , no motion, no fire), the Firestore status field reads "Safe", the relay remains de-energised, and the mobile app displays a green status indicator.

2) Gas Leakage Simulation: Incrementing the MQ-2 analogue input above 3800 in the Wokwi simulator immediately updates the Firestore status to "Danger", activates the relay and buzzer, and triggers a red alert on the mobile dashboard within one second.

3) Temperature Exceedance: Setting the DHT22 temperature input above 50°C similarly transitions the system to the Danger state, demonstrating correct multi-parameter threshold evaluation.

4) Motion Detection: Toggling the PIR sensor HIGH in simulation activates all actuators and pushes a Danger status to Firestore, confirming unauthorised-access detection.

D. Discussion

The experimental results confirm that the proposed system meets its design objectives: real-time multi-parameter monitoring, autonomous local actuation, and instantaneous cloud-based remote alerting. The sub-1.5-second end-to-end latency is acceptable for laboratory safety applications where human response time is typically several seconds.

Limitations include dependence on stable Wi-Fi connectivity; a network disruption would prevent cloud updates while local actuation (relay, buzzer) remains functional due to onboard threshold logic. Future work will explore offline buffering via ESP32 non-volatile storage (NVS) and MQTT-based communication as a lower-latency alternative to the Firestore REST/SDK channel.

VII. CONCLUSION

This paper presented a low-cost, IoT-based Smart Laboratory Safety System employing an ESP32 microcontroller, DHT22, MQ-2, PIR, and fire sensors for continuous multi-parameter environmental monitoring. The system performs autonomous hazard response through a relay and buzzer while simultaneously synchronising data to a Firestore Realtime Database. A companion Android application, developed with MIT App Inventor, provides authenticated remote monitoring with real-time colour-coded safety status indication.

Experimental validation through Wokwi simulation confirmed sub-1.5-second end-to-end alert latency and robust threshold-based hazard detection across all tested scenarios. The system offers a scalable, replicable framework suitable for academic laboratories, chemical storage areas, and any environment requiring real-time safety supervision.

Future enhancements will include: (i) integration of a flame sensor replacing the push-button simulation, (ii) SMS/email emergency notification via Twilio or SendGrid, (iii) historical data analytics and trend visualisation, and (iv) edge AI inference for anomaly detection independent of cloud connectivity.

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